HVAC Control

Dashboard Filters

Select Day

All Days

System Information

Data Source: Synthetic IoT Data

Sampling: 1 hour

Monitoring: HVAC System

Mode: Analytics

HVAC Energy Optimization

IoT-Based HVAC Monitoring & Energy Analytics Dashboard

Live System Overview

Temperature	Humidity	Energy	Occupancy	Est. Cost
24.5 °C	61.4 %	631.2 …	38	₹5049

HVAC System Status

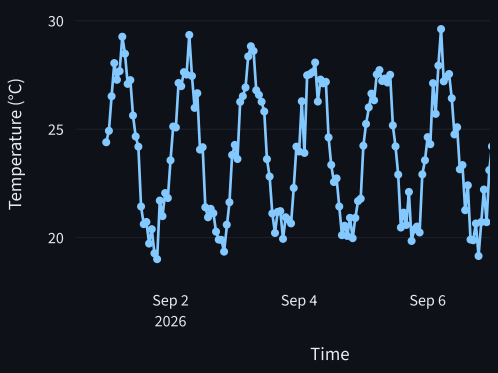
HVAC SYSTEM: ON

Current Power: 3.86 kW

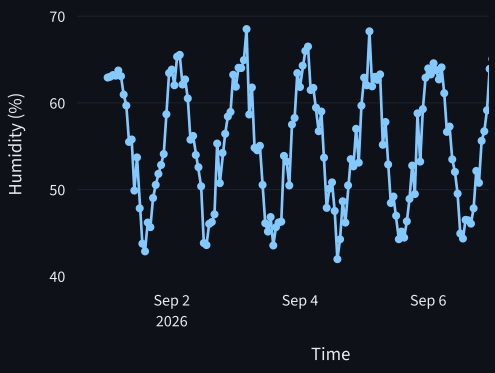
Peak Energy: 5.30 kWh

Environmental Monitoring

Temperature Trend

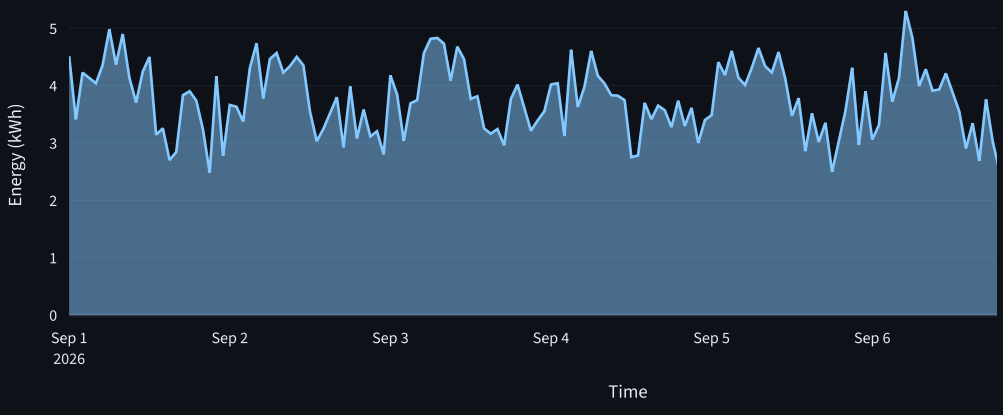


Humidity Trend



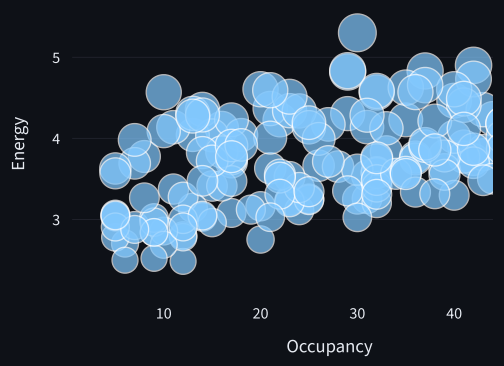
Energy Analytics

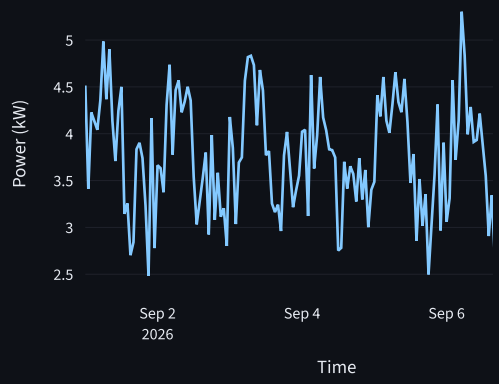
HVAC Energy Consumption



Occupancy vs HVAC Energy

HVAC Power Consumption





Energy Performance

Average Temperature	Average Humidity	Peak Energy	Potential Saving
24.0 °C	55.2 %	5.30 kWh	15.3%

Smart Optimization

HVAC operation is currently within the expected monitoring conditions.

System Alerts

High temperature detected: 29.6 °C

High energy consumption detected: 5.30 kWh

IoT Sensor Data

View Raw Sensor Data

Time	Temperature	Humidity	Occupancy	HVAC Power
2026-09-01 00:00:00	24.3974	62.9239	47	4.5144
2026-09-01 01:00:00	24.9247	63.0189	15	3.4129
2026-09-01 02:00:00	26.5182	63.2098	22	4.227
2026-09-01 03:00:00	28.0469	63.139	16	4.1278
2026-09-01 04:00:00	27.2768	63.7323	13	4.0419
2026-09-01 05:00:00	27.6764	63.0791	14	4.3575
2026-09-01 06:00:00	29.2634	60.9564	48	4.9841
2026-09-01 07:00:00	28.4777	59.6954	21	4.3684
2026-09-01 08:00:00	27.0885	55.4978	42	4.9009
2026-09-01 09:00:00	27.2625	55.7775	11	4.1382

HVAC Energy Optimization | IoT Analytics | Python + Streamlit | Synthetic Sensor Data